

Board Independence and Liquidity:

Evidence from East Asia

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Abstract

Using a sample of East Asia companies, we investigate the link between board composition and the liquidity of a firm's shares. On average, greater board independence significantly increases liquidity. However, that relation is mediated by several other variables that influence the flow of information from managers to the board and the public. The beneficial impact of board independence on liquidity is reduced when more sophisticated brokers make a market in shares, when board-management relations are strained, or when CEOs have greater bargaining power relative to the board. Legal environments that provide greater investor protections also tend to attenuate the impact of board independence. Our findings challenge the notion that restrictions on managers necessarily enhance informational environment and liquidity.

Key Words: board composition, liquidity, corporate governance

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1. Introduction

Regulators, financial economists and accountants have long viewed independent directors as contributing to effective corporate governance. Firms with more independent directors generally exhibit higher profitability and greater discipline on executive compensation (Black and Kim, 2012; Liu et al., 2015; Weisbach, 1988). However, board independence comes with its own challenges. Boards may facilitate corporate governance by advising as well as disciplining managers, and good advice and effective monitoring require a framework of trust and information sharing. When relations with their boards are strained, CEOs may actively attempt to suppress the information reaching the board, especially when it is dominated by independent directors (Hermalin and Weisbach, 2012; Armstrong et al., 2014; Mueller and Inderst, 2005; Adams and Ferreira, 2007).¹

In particular, the CEO's control over information may limit the independent board's ability to effectively monitor and advise, which could in turn degrade the strength and speed of deliberation and decision making. As a general rule, the net benefit of board independence may be contingent on potential negotiations between the board and the CEO, or more generally, on the way a CEO deals with the board. While independent boards may better align the interests of the CEO with outsiders in some circumstances, they may also generate higher tension within the company, creating a wedge between the interests of the CEO and outside investors. The value-impact of the relation between a CEO and an independent board is still unproven in the literature, and may in fact be conditional on other firm characteristics.

We attempt to measure some of these trade-offs using stock-market liquidity as a proxy for the firm's information environment. Market makers and other traders are less willing to supply liquidity when exposed to adverse information events.² Therefore, liquidity will be greater when information events are less likely, which in turn may depend on the openness of the relation between managers and the board. Our view of the role of the market maker builds on the conventional modeling of liquidity as a response to asymmetric

¹ Existing research shows that where CEOs face significant career risk, they may be particularly reluctant to share information with independent boards. For example, Mueller and Inderst (2005) study a model in which the CEO can hide information that would allow the independent board to conclude that he should be replaced.

² For example, evidence in Doukas et al. (2004) and Doukas and McKnight (2005) implies that uncertainty can affect liquidity. Coughenour and Saad (2004) show that liquidity provided by market makers depends on the risk of providing it. Comerton-Forde et al. (2010) find that liquidity can worsen when market makers lose money from adverse information events.

information. Our focus is on market maker confidence, which may itself be a function of governance arrangements. Thus, the board's structure may influence, for better or for worse, the information environment, market makers' confidence, and ultimately liquidity.

We focus on variation in board structure and liquidity across a sample of East Asian firms. This is an interesting sample both because of the role of liquidity in the East Asian financial crisis of 1997-1998 (O'Hara, 2003; Fernando and Herring, 2003; Coughenour and Saad, 2004) and because of the regulatory response it elicited. In the wake of that crisis, several countries implemented new legislation calling for an increase in the representation of outside directors on boards. For example, in 2001 Korea implemented governance rules requiring large firms to have 50% outside directors and small firms to have 25% outside directors (Black and Kim, 2012). In 2001, the Kuala Lumpur stock exchange required one-third of the board of directors of all listed companies to be independent, with a minimum of two independent directors. Starting in 2007, Taiwanese public firms were required to appoint at least two independent directors (Chang and Wei, 2011). Therefore, our results provide important context for the debate over the costs and benefits of the regulations that emerged from that crisis, especially the debate focusing on reform of corporate governance.

This paper complements recent research examining the economic consequences of such regulations. We argue that the impact of greater informational transparency may be contingent on the negotiating strength of the CEO versus the board. Such negotiations may be cumbersome in the absence of the trust that can be established among a group of insiders. Therefore, while our empirical analysis begins with the average effect of board independence on stock market liquidity, our focus is on the interaction effects of other variables with board independence. In particular, we ask whether market makers believe that more independent boards are viewed as mitigating information risk, whether their trading behavior responds to board composition, and, importantly, whether other variables affect the relation between board composition and liquidity. We hand collect governance data for a sample comprising 2,407 firm-year observations over the period 2004-2013 in East Asian stock markets. Our measures of liquidity include Amihud illiquidity, Roll spread, zero returns, and the effective bid-ask spread. We employ the dynamic panel generalized method of moments (GMM) estimator to control for dynamic endogeneity, unobservable heterogeneity, and simultaneity.

As in earlier studies (e.g., Armstrong et al., 2014; Kanagaretnam et al., 2007; Fama and Jensen, 1983), we find that the proportion of independent directors is positively related to liquidity. However, the strength of this relation significantly interacts with several other variables. For example, the beneficial impact of board independence on liquidity appears less pronounced when: (a) market makers are better equipped to assess numerous other sources of information; (b) the board is engaged in considerations that threaten the job of the current CEO; (c) the CEO has greater negotiating power relative to the board; and (d) there are greater legal protections for investors that make board supervision of managers less salient. Our empirical estimates suggest that these mediating variables can negate more than half of the typically beneficial effects of independent boards on liquidity.

This work advances our understanding of the importance of the communication structure among a group of insiders and contributes to the literature in several ways. First, we show that the relative benefits and costs of increasing board independence vary across firms and over time, responding to the negotiating strength of the CEO versus the board. Our empirical results raise the concern that in some circumstances the CEO will suppress the information reaching the board as a way to entrench himself, particularly when the board is more independent. Therefore, the relative advantage of board independence must be conditioned on both the negotiating strength of the CEO relative to the board as well as on the information environment. We also extend the literature on firm disclosure. We present evidence that more effective and sophisticated market makers can almost fully substitute for the liquidity-enhancing informational effects otherwise generated by independent boards. Finally, we contribute to the literature on the determinants of board structure with respect to investor protections provided by the legal system in the firm's home country. We find that the beneficial impact of independent boards is greater when legal protections for investors are weaker. In the next section of the paper, we briefly review the literature and develop our hypotheses. Section 3 presents our data and Section 4 contains our empirical analysis. Finally, Section 5 concludes.

2. Literature and Hypothesis Development

2.1 *Information versus cooperation*

Since the East Asia liquidity crisis, liquidity and governance have been major policy concerns in these countries. However, as we will argue below, the relation between board composition and liquidity is not theoretically clear, and the empirical evidence on the link between stock market liquidity and board characteristics such as independence is not definitive. One of the key points we raise is that the information environment depends on information sharing and the framework of trust between the CEO and the board. Thus, an independent board should be viewed as a two-edged sword: Increasing monitoring by an independent board may be helpful in mitigating informational asymmetries when it leads to greater information disclosure, but it can also generate its own agency costs by aggravating incentives for managers to manipulate the quality of information. This point builds on two groups of studies that have investigated the effect of board independence on the informational environment. Fama and Jensen (1983), Adams and Ferreira (2007), Kanagaretnam et al. (2007), Ajinkya et al. (2005), Goh et al. (2014) and Armstrong et al. (2014) are the main advocates of the point of view that board independence increases disclosure. They believe that independent directors demand a substantial amount of firm-specific information to effectively perform their advising and monitoring duties. Conversely, inside directors recognized as active traders in their company's stock earn substantial abnormal returns (Mishra, 2011; Chung and Charenwong, 1998), implying that heavy inside representation is associated with greater use of inside information. On balance, therefore, independent director domination of the board appears to encourage information disclosure.

However, another group of studies suggests that monitoring by an independent board also entails costs by encouraging opportunistic CEO behavior (Adams and Ferreira, 2007; Charoenwong et al., 2011; Hermalin and Weisbach, 2012; Bazrafshan, Kandelousi and Hooy, 2015). These studies argue that considerable relevant information is under the CEO's control, and that the CEO has an incentive to manipulate information, for example, to present the current situation of the firm in the most favorable light. Information is one way that the CEO can wield a strong influence over the board's decisions and entrench his position. Outside directors are presumably a greater threat to a CEO's career than inside directors, and thus present a greater incentive to manipulate disclosures. Moreover, independent directors typically have comparatively smaller equity stakes in the firms they monitor, and so have

weaker financial incentives to carefully act as agents of shareholders or to uncover unfavorable information (Guo and Masulis, 2012).

2.2 *Board Structure and Liquidity*

To study the relative importance of these competing considerations, we employ interaction variables to investigate how the impact of board independence on stock market liquidity varies with several other governance variables, specifically, CEO bargaining power, CEO career risk, and legal protections already in place. The goal is to test whether these variables mediate the impact of the CEO-independent board relation and ultimately the information environment in which shares are traded.

While board independence can affect stock market liquidity through general information channels concerning the transparency of the firm, it can also have a more direct effect through the impact of information on market makers. A market maker trades off an exposure to losses from trading with informed agents against the prospect of profits on liquidity-motivated transactions. As the market microstructure literature emphasizes, that tradeoff and the liquidity that market makers are willing to provide depend on both information availability and asymmetry (e.g., Garman, 1976; Stoll, 1978; O'Hara and Oldfield, 1986; Chordia et al., 2000; Hasbrouck and Seppi, 2001; Coughenour and Saad, 2004; Comerton-Forde et al., 2010).

One factor affecting the impact of board structure on liquidity is market maker sophistication. One would expect to observe meaningful variation across market makers in the ability to ferret out relevant information rather than relying solely or even primarily on firm disclosures. We will use the existence of high-frequency trading capabilities as a proxy for market maker sophistication. For example, in theoretical models by Brogaard et al. (2014); Chordia et al. (2013) and Chaboud et al. (2014), high-frequency trading market makers (HFT market makers) are classified as informed traders, who have a distinct advantage over traditional market makers in detecting and reacting more quickly to new information. Their algorithmic trading programs allow them to quickly process large amounts of information contained in the order flow and price movements. Kirilenko et al. (2014) also show that HFT traders are highly informed: on average, they tend to buy as prices are about to increase and tend to sell as prices are about to decrease. Therefore, we will treat HFT

activities as a proxy for the differential abilities of market makers to value firms and ask whether that analytic ability affects liquidity and the market reaction to the flow of information available from the firm.

Another mediating factor is potential tension between the board and management. Independent boards are recognized as particularly effective in performing specific tasks, such as hiring and firing the CEO (Weisbach, 1988), adopting takeover defenses (Brickley, Coles and Terry, 1994), and negotiating takeovers (Cotter, Shivdasani and Zenner, 1997),³ all circumstances in which CEO job security is threatened. Cornelli and Karakis (2015) find that percentage of outsiders on the board during LBOs similarly increases the probability of CEO turnover. While CEOs may generally have qualms about sharing information with independent directors, these reservations will be heightened when their jobs are at risk, thus putting a limit on the board's ability to effectively monitor. Therefore, CEO career risk may reduce information sharing between managers and boards and affect the net benefit of an independent board.

Several papers conclude that CEO incentives to limit the effectiveness of director's monitoring can elicit actions aimed at signal distortion. Hermalin and Weisbach (2012) show that when managers are concerned about job security, they change their investment strategy toward myopic behavior to improve reported performance, for example, substituting away from longer-term investments such as R&D, toward shorter-term investments or actions that affect reported profitability sooner. Charoenwong et al. (2011) argue that when subjected to greater scrutiny, CEOs direct their efforts toward increasing their firms' stock prices by managing or massaging information in order to prevent a takeover. Overall, CEO turnover may be a signal about the presence of tense negotiations inside the company and the potential for greater opacity concerning the firm. To explore this possibility, we examine changes in the impact of board independence during periods in which the board is engaged in considerations that threaten CEO job security.

Higher-performing CEOs, who are harder to replace, will generally have greater bargaining power and be better positioned to negotiate with the board. Because CEO skill and success should be associated with higher values for Tobin's Q, high-Q firms may tend to

³ Kaplan and Minton (2011) show that board-driven turnover rises significantly, increasing from 10.9% during the 1992 to 1999 period to 12.4% in the 2000 to 2007 period of the sample.

be ones in which the board has less power relative to the CEO. For example, Hermalin and Weisbach (1998) indicate that the intensity with which the board monitors the CEO decreases with its prior estimate of the CEO's ability. Therefore, we will use Tobin's Q as a proxy for the differential bargaining power of the CEO to control board activities and estimate how bargaining within the company affects the net benefit of an independent board.

Finally, the legal environment may potentially affect the impact of board independence (Chung et al., 2012). High levels of investor protection may provide a substitute for transparency and reduce the importance of independent directors. For example, Klapper and Love (2004) show that investor protection in corporate governance matters more in countries with weak investor protections. However, it is also plausible that legal protections are complementary to board independence, aiding boards in information production and dissemination, and thus increasing the beneficial impact of independent directors. Moreover, strong external protections that restrict executive prerogatives might take the pressure off the board to curtail management discretion and therefore remove a source of friction between them (Chung et al., 2012; Claessens et al., 2000; Lombardo and Pagano, 2006; La Porta et al., 1999). Therefore, we will consider proxies for the legal environment and estimate how that environment affects the impact of an independent board.

2.3 *Hypotheses*

These considerations give rise to the following hypotheses. Our underlying premise is that a more transparent environment reduces market makers' vulnerability to information shocks and thereby increases liquidity. Independent boards may increase pressure on firms for disclosure and transparency, but may also increase CEO incentives to hide information; therefore, Hypothesis 1 is two sided:

Hypothesis 1. Liquidity varies with the representation of independent directors on the board.

Our primary focus, however, is on other factors that can interact with board independence. For example, market makers with greater access to market knowledge may be better equipped to process information about the firm, and be better able to assess the quality of the information the firm releases to the public. Their trading strategies may also allow for

more sophisticated and effective risk management. These higher-resource firms will be less dependent on informational transparency when assessing their vulnerability to information events. Therefore, a second hypothesis is that:

Hypothesis 2. The impact of board independence on share liquidity is lower when the analytic capability of firms making a market in those shares is higher.

In some circumstances, particularly when managers are concerned for their job security, we have noted that more independent boards may be more threatening to managers, which can induce management to impede the flow of information. CEO attempts to manipulate information also have other side consequences. First, they reduce information sharing between managers and boards and may cause a break in communication between CEOs and directors (Adams and Ferreira, 2007). The CEO may therefore lose the expertise of board members that would otherwise be available. Second, they may also erode the strength and speed of deliberation and decision-making (Faleye et al., 2011). In these periods, the benefit of board independence will be reduced. Therefore, a third hypothesis is:

Hypothesis 3. The impact of board independence on share liquidity is less positive in periods of board-instigated CEO turnover.

Because the ability of the board to affect information transparency depends on the negotiating power of management, the influence of independent directors may be reduced when CEOs have greater bargaining power. When better-performing CEOs have greater leverage, we would expect board-driven improvement in corporate transparency to be muted. In the liquidity context, market makers would rationally limit trading exposure or demand higher compensation for inventory risk in these circumstances, as they react to the increased potential for adverse selection. Allowing Tobin's Q to proxy for management's negotiating power suggests the following hypothesis:

Hypothesis 4. The positive impact of board independence on share liquidity is lower in firms for which Tobin's Q is higher.

Finally, the legal environment may potentially affect the impact of board independence (e.g., Chung, et al., 2012). High levels of investor protection may provide a substitute for transparency and reduce the importance of independent directors. However, it is also possible that legal protections are complementary to board independence, aiding boards in information production and dissemination, and thus increasing the beneficial impact of independent directors. Moreover, strong external protections that restrict executive prerogatives might take the pressure off the board to curtail management discretion and therefore remove a source of friction between them. Therefore, a two-sided hypothesis is that:

Hypothesis 5: The impact of board independence on share liquidity depends on the degree of investor protection provided by the legal system in the firm's home country.

3. Data and Research Design

3.1. Sample and Data

Our sample comprises the market-listed companies on East Asia's nine main stock exchanges (China, Hong Kong, Japan, Malaysia, Singapore, South Korea, Taiwan, Thailand, and Indonesia). The choice of these countries is dictated by the accessibility of their annual reports within the East Asian region; moreover, these countries all passed mandatory regulation reforms to increase independent directors on the boards of listed companies. Our sample covers the period from 2003 to 2013, and contains 2,407 firm-year observations. Financial statement data are taken from Compustat, which provides data for East Asia stock companies from 2003. Consistent with prior research (e.g., Linck et al., 2008; Duchin et al., 2010), we define a director to be independent if he or she is identified in the annual report as an outsider who is not employed by the firm and has no material economic interest in its success.

We employ four different measures of illiquidity: Roll's (1984) implicit spread, the proportion of zero daily returns (Lesmond, Ogden and Trzcinka, 1999), effective bid-ask spread, and Amihud's (2002) measure. This variety of measures is motivated by the fact that a single measure cannot capture all dimensions of liquidity. Each of these liquidity measures

has advantages and disadvantages, but we tend to find similar results regardless of which measure we use, suggesting that our findings are not due to issues specific to any particular measure. These liquidity measures are all measured at low frequency. In Section 3.2, we present both theoretical as well as empirical arguments suggesting that low frequency measures are appropriate for this study.

We use as controls factors that are generally accepted as determining liquidity (e.g., Armstrong et al., 2014; Chung et al., 2010; Jain et al., 2008; Stoll, 2000): trading volume (mean daily dollar-denominated trading volume), the leverage ratio (debt divided by book assets), firm size (book value of assets), and return volatility (standard deviation of daily closing quote-midpoint returns). In accordance with the empirical evidence, we predict that higher trading volume and firm size will lead to an increase in liquidity while greater leverage and return volatility will reduce it.

Market making is now often carried out by high-frequency proprietary algorithms. As in other markets, East Asian stock markets consist of three “trader types,” institutions, individuals, and program traders. The last trader type includes HFTs that effectively function as market makers and whose trading strategies may be different from those of other traders. Following O’Hara (2015) and O’Hara et al. (2014), we classify two firms making markets in East Asian companies, Barclays Capital and Goldman Sachs, as sophisticated HFT market makers. They note that the activity of these two firms corresponds well to the definition of high-frequency trading in the SEC Concept Release on Equity Market Structure (2010).⁴ To compile a list of companies whose stock is traded by these firms, we use their stock inventories. These data allow us to investigate the role some high-frequency traders play in liquidity provision.

As noted, the threat of CEO turnover can affect the extent to which board independence contributes to making information available to the market. Therefore, we hand-collect data on CEO turnover episodes to find periods in which this threat is most manifest.⁵ We then search annual reports for the exact turnover announcement date and,

⁴ The Securities and Exchange Commission's (SEC's) Concept release on equity market structure (2010) defines them as “professional traders acting in a proprietary capacity that engage in strategies that generate a large number of trades on a daily basis.”

⁵ Related studies have recommended hand collection of data on CEO turnover. They believe conventional data sources are not sufficient to identify periods of potential CEO turnover because they list several officers for each firm and do not list them in order of importance. Moreover, for our study in particular, most data sources are not

following Jenter and Kanaan (2015), classify each CEO turnover according to whether the turnover was board driven (forced) or voluntary. All departures for which the annual reports mention that the CEO was fired, forced out, or retired or resigned due to policy differences or pressure are classified as board driven. All other departures for CEOs aged 60 or older are classified as voluntary. Departures for CEOs below age 60 are also classified as voluntary if the annual report mentions the reason as death, poor health, or the acceptance of another position (including chairmanship of the board). Following Defond and Hung (2004), the turnover year is the year in which a new name appears as chief executive.⁶ The years with turnovers are treated as ones in which board-CEO relations are most likely to be strained.

We refer to the firm's top corporate officer as the "CEO," but the title used for a firm's top manager can differ across countries. Moreover, even firms within a country do not necessarily consistently use the same title to identify the top manager. Given that top executive titles vary both across and within countries, we identify CEOs using the following procedures: first, we identify whether companies have executives with the title chief executive officer, chief executive, or CEO.⁷ Second, for companies without officers with the preceding titles, we search the annual report to identify the person with another title such as chief executive, president, general manager, or managing director who nevertheless has CEO power. We identify the CEO as the person with the title chief executive officer or president for Taiwanese, South Korean, Thai, or Japanese companies. For Singaporean, Indonesian, Chinese, Hong Kong and Malaysian companies, the CEO may have the title of chief executive officer or managing director. If two managers have the same title, indicating shared responsibility, we drop that firm-year from the sample to avoid dealing with split turnover.

3.2 *Liquidity Measures*

Liquidity embodies both direct and indirect costs of trading, some of which are best measured at high (intraday) frequency and others at lower frequency. A direct trading cost is

consistent with the placement of family names for most East Asian countries, in which the sequence of first name and last name differs from the Western convention (Defond and Hung, 2004).

⁶ For example, if the CEO in 2012 is Smith and the CEO in 2013 is Jones, we classify 2013 as the turnover year.

⁷ We exclude the title deputy chief executive in identifying the top executive officer.

the bid-ask spread while an indirect trading cost includes price-impact measures. However, there are several serious obstacles to use of intraday liquidity measures in this study. Trading data in emerging markets are of relatively limited quality, and detailed transaction prices are not widely available. We use a time period that predates the general availability of intraday data, and indeed, such data are still not available for some East Asia markets. This study requires a relatively long time-series sample, because it relies on within-country or within-firm variation in the corporate governance environment. Therefore, we will rely on lower-frequency measures. Corwin and Schultz (2012) argue that these measures are informative in a wide variety of applications. Goyenko, Holden and Trzcinka (2009) show that these proxies successfully measure liquidity in stock markets.

The first measure of liquidity is Roll's spread, which uses the serial covariance of price changes as an estimate of the spread. Roll (1984) assumes that the true value of a stock follows a random walk and that P_t , the observed closing price on day t , is equal to the stock's true value plus or minus half of the effective spread. Under these conditions, the expected autocorrelation of returns from observed prices is negative, and Roll derives the following simple estimator for the effective spread: $2 \sqrt{-\text{Cov}(\Delta P_t, \Delta P_{t-1})}$

We will use Goyenko, Holden and Trzcinka's (2009) modified version of Roll's estimator, as expressed in the following equation:

$$\text{Spread} = \begin{cases} 2\sqrt{-\text{Cov}(\Delta P_t, \Delta P_{t-1})} & \text{when } \text{Cov}(\Delta P_t, \Delta P_{t-1}) < 0 \\ 0 & \text{when } \text{Cov}(\Delta P_t, \Delta P_{t-1}) \geq 0 \end{cases}$$

For each stock, we compute the average Roll spread in each month.

The second measure of liquidity is the proportion of zero daily returns, suggested by Lesmond, Ogden and Trzcinka (1999). We follow Bekaert et al. (2005) to construct this proxy. Goyenko et al. (2005) compare various transaction cost measures for U.S. data, and find that the observed zero return measure is correlated with effective costs obtained from higher frequency data. Two arguments support this measure. First, stocks with lower liquidity are more likely to have zero volume days and thus are more likely to have zero reported-return days. Second, as Lesmond, Ogden and Trzcinka (1999) argue, stocks with

higher transaction costs elicit less private information acquisition (because it is more difficult for the value of information uncovered to overcome higher transaction costs). If the value of an information signal is insufficient to outweigh the transaction costs, market participants will elect not to trade, resulting in an observed zero return. Thus, a high proportion of zero returns is symptomatic of high transaction costs. The use of this liquidity proxy also allows us to keep substantially more observations than other proxies for liquidity. The resulting monthly measure of liquidity, zeros, is simply:

$$\text{Zeros} = \frac{\text{Days in month with zero returns}}{\text{Trading days in month}}$$

Our third liquidity measure is the effective bid-ask spread. Stoll (1978) and Glosten and Harris (1988) show that bid-ask spreads reflect the adverse selection cost that market makers incur from trading with better informed investors. As in Stoll (1978), we measure the relative quoted bid-ask spread as the difference between the bid and ask prices divided by their average:

$$\text{Effective bid-ask spread} = \frac{\text{Ask price} - \text{bid price}}{(\text{Ask price} + \text{bid price})/2}$$

The final liquidity measure is based on Amihud (2002). This measure captures how much a given trading volume moves prices. It has the intuitive interpretation of an estimate of price impact, consistent with the notion that the price pressure associated with buying and selling reflects a stock's liquidity (Grossman and Miller, 1988). Goyenko et al. (2009), who compare many popular low frequency bid-ask spread and price-impact proxies with their high frequency counterparts, find the lower frequency Amihud measure is the best measure for price impact. It is calculated as the absolute daily return on the stock, R_t , divided by the firm's daily dollar volume, averaged over the month. The average is calculated over all positive volume days, since the ratio is undefined for zero volume days.

$$\text{Amihud illiquidity} = \text{Average} \left(\frac{|R_t|}{\text{Volume}_t} \right)$$

Because the Amihud measure uses a local currency unit to measure trading volume, we convert trading volume to U.S. dollars to make this measure comparable across firms in different countries.

3.3 *Research Design*

Endogeneity is a serious concern when carrying out any empirical estimation in board structure (Lehn et al., 2009; Harris and Raviv, 2008; Hermalin and Weisbach, 2003; Wintoki, Linck and Netter, 2012). It is plausible that board structure changes are determined by past, present, and even expected future characteristics of the firm. In addition, the current values of many governance variables are likely to depend on past values of themselves and other variables. Thus, neglecting this source of “dynamic endogeneity” can have serious consequences for inference.

To alleviate potential biases that may arise in this context, we employ dynamic panel GMM estimated using three different techniques to control for simultaneity, dynamic endogeneity, and unobserved heterogeneity.⁸ We circumvent the potential problems associated with the effect of simultaneity by using instrumental variables. The advantage of using a GMM estimator is that we can treat endogeneity without the need to identify external instruments. Instead, GMM uses information from a firm's history as instruments for the current values of liquidity, board structure, and other explanatory variables or firm characteristics.⁹

Wintoki et al. (2012) caution that when we use a combination of variables from the firm's history as instruments to account for simultaneity, we must ensure that the underlying economic process is itself dynamic. In our context, this requires that current governance is related to past liquidity. Therefore, we will rely on a set of internal instruments contained within the panel itself: past values of board structure and liquidity can be used as instruments

⁸ Endogeneity in the relation between independent directors and liquidity can result from: unobservable heterogeneity (which arises if there are unobservable factors that affect both the dependent and explanatory variables); simultaneity (which arises if the independent variables are a function of the dependent variable or expected values of the dependent variable); and dynamic endogeneity (for example, if the current liquidity of a firm leads the firm to take actions that affect the future independence of the board and therefore the firm's future liquidity).

⁹ Increasing the instruments' lag length makes them “more exogenous,” but may also make them weaker. In our specifications, we use lag one and present tests that rule out potential endogeneity. Wintoki, Linck & Netter (2012) believe that even given the occasional weakness of the specification tests accompanying the dynamic GMM estimator, it likely still dominates inferences from OLS or fixed-effects estimation if the underlying economic process is dynamic.

for current realizations of board structure. Building on prior research, we argue that dynamic endogeneity is of concern in independent board and liquidity analysis. The relevant literature has evolved along two major lines. Some studies (Bhide, 1993; Linck et al., 2008; Lehn et al., 2009; Duchin et al., 2010) have demonstrated that stock liquidity affects corporate transparency, information processing costs, and the opportunity cost of outside directors, all of which are factors that may affect future optimal board structure. The relation between current corporate transparency and the future proportion of independent directors suggests that firms populate their future boards with a high proportion of independent directors only when the information environment will be sufficiently transparent to allow independent directors to effectively perform their monitoring and advising duties (Linck et al., 2008; Lehn et al., 2009).

A second argument in support of dynamic GMM builds on insights from work by Fang et al. (2009) and Wintoki et al. (2012), who show that the presence of independent directors on the board is partly a function of past firm performance. If the CEO's performance is also related to the liquidity of the stock, then past liquidity can be correlated with current board independence. Wintoki et al. (2012) and Pathan and Faff (2013) also provide strong evidence that the instruments associated with a dynamic GMM approach are valid and powerful, and particularly useful when it is difficult to find orthogonal instruments to reduce the endogeneity problem in governance variables such as board independence.

Further, we adopt a dynamic panel GMM estimation procedure that is robust to "dynamic endogeneity," i.e., the potential for a firm's current liquidity to affect both its future liquidity as well as the composition of its board. We use a fixed-effects panel model in our standard regression specifications to produce consistent parameter estimates robust to unobservable heterogeneity (Petersen, 2009). Driscoll and Kraay (1998) show that ignoring cross-sectional dependence in the estimation of linear panel models can lead to severe biases.

The dynamic GMM approach that we employ is preferable to standard fixed effects estimation for two reasons. First, Wintoki, Linck and Netter (2012) show that standard fixed-effects estimators are biased when the dynamic relations between the variable of interest and the explanatory variables are important. They note that "traditional fixed-effects estimation can potentially ameliorate the bias arising from unobservable heterogeneity. However, it does this at the expense of a strong exogeneity assumption, one that is often not explicitly

recognized by researchers. That is, it assumes that current observations of the explanatory variable are completely independent of past values of the dependent variable” (p.582). In our application, we argue that current liquidity affects future characteristics of the board. Therefore, we cannot use standard fixed effects. The second advantage of the GMM approach is that it alleviates the problem of slow moving corporate governance indices over time, which could render fixed-effects techniques ineffective (Hoechle et al., 2012; Coles et al., 2012).

We therefore conduct a dynamic panel two-step GMM regression following Wintoki, Linck and Netter (2012) to control for potential dynamic endogeneity.¹⁰ The reliability of the system GMM estimates is also checked using Hansen’s test for instrument validity and Arellano and Bond’s (1991) test for serially uncorrelated error terms¹¹.

The lower panels of Tables 2–5 include some tests for the reliability of the system GMM estimates. AR (1) and AR (2) are Arellano and Bond (1991) statistics for first-order and second-order autocorrelation in the first difference errors. The AR (1) and AR (2) tests show no evidence of autocorrelation at conventional levels of significance for any of the specifications in these tables. The Hansen J-test of over-identifying restrictions is a test with the joint null hypothesis that instrumental variables are valid (i.e., uncorrelated with error terms). These tests of exogeneity yield *p*-values not even close to statistical significance in all of Tables 2-5.

One potential danger of system GMM implementation is instrument proliferation (Roodman, 2009). Instrument proliferation in the system GMM estimator may bias the estimated coefficients towards those from non-IV estimators, such as OLS, and severely weaken the power of the Hansen J test to detect invalid instruments. To mitigate instrument proliferation, we use techniques suggested by Roodman (2009) to reduce the instrument count. The reported instruments and clusters count in the tables suggest that instrument

¹⁰ We use the `xtabond2` command in STATA for the GMM estimation. In the system GMM, two-step estimates of the standard errors tend to be downward biased (Blundell & Bond, 1998) and hence a finite sample correction to the two-step covariance matrix is executed which uses optimal weighting matrices (Windmeijer, 2005).

¹¹ Since we cannot rely on the Sargan statistic as a model specification test for the GMM estimation (due to the presence of heteroskedasticity, detected using a test proposed by Pagan and Hall (1983)), we justify the validity of the instruments using a test of second-order correlation. If we include enough lags to control for the dynamic aspects of our empirical relationship and if the assumptions of our specification are valid, then the residuals in the first differences, AR(1), should be correlated, but there should be no serial correlation in second differences, AR(2).

proliferation is unlikely to be a problem in our estimation. Overall, these post-estimation results indicate that the dynamic model is a reasonably good specification for the liquidity model.

4. Results

4.1 Descriptive Statistics

Table 1 reports summary statistics for the sample. Panel A shows statistics for the full sample, while Panel B reports country-specific summary statistics. Liquidity varies enormously across this sample of East-Asian firms, regardless of which liquidity measure is used. For most liquidity measures, standard deviations across firms are multiples of the mean. The two elite market makers support shares of only a small fraction (6.7 percent) of the firms in the sample. The mean value for CEO turnover indicates that board-driven CEO turnover was experienced by 10% of our sample. The maximum value of the percentage of independent directors is 100% and the minimum value is 3.2%. The independent board variable has an average value of 39.0%. The minimum logarithm value of firm size is 7.82 (or \$2,492,000) and the maximum logarithm value is 21.66 (or \$2,568,690,941,000). An average of 23% of the companies' assets is financed with debt.

[Table 1 here]

Table 2 presents the Pearson and Spearman rank of correlations between the liquidity measures, independent board and all control variables used in our baseline specifications. Pearson correlations are reported above the main diagonal and Spearman correlations are reported below the diagonal. The various illiquidity measures generally demonstrate positive and highly significant correlations. Both Pearson and Spearman correlation coefficients indicate a generally significant negative association between board independence (IndepBoard) and illiquidity. There is significant correlation between the majority of control variables and proxies for liquidity. We also check for multicollinearity statistically by calculating the condition index.¹² Though not tabulated, we find that multicollinearity is of little concern.

¹² The condition index is the square root of the maximum eigenvalue divided by the minimum eigenvalue. If this index is more than 30, the data exhibits a severe multicollinearity problem (Gujarati, 2004).

[Table 2 here]

4.2 *Relation between Board Independence and Liquidity*

Table 3 shows basic results for the impact of board independence on liquidity. Each of the four columns corresponds to a different liquidity measure, and because each measure actually measures illiquidity, the consistently negative and statistically significant coefficients indicate that board independence predicts greater stock market liquidity. Thus, at least on average, board independence appears to contribute to liquidity.

[Table 3 here]

Coefficients on the other control variables in Table 3 are generally consistent with those in Armstrong et al. (2014) and Chung et al. (2010). The coefficients on firm size are all negative, and statistically significant at 1% or better in two of the four regressions, indicating that large firms have greater liquidity. When statistically significant, the coefficients on trading volume are also negative. Similarly, volatility is generally insignificant, but when it is significant, its coefficient is positive, implying that return variation reduces liquidity. The leverage term is positive, but significant in only one specification.

4.2.1 *HFT Market Makers*

In Table 4, we ask whether the effect of board independence depends on the presence of HFT market makers by introducing an interaction term between *IndepBoard* and an indicator variable for coverage by HFT market makers. The results strongly suggest that board independence has less impact on liquidity when these elite brokers make a market in the firm's shares. The net effect of board independence on share liquidity in the presence of HFT market makers is given by the sum of the coefficients on *IndepBoard* and *IndepBoard* $\times$ *HFT-MM*. For all but the Zeros measure of liquidity, while the coefficient on *IndepBoard* is negative (implying that liquidity improves with independent directors), the coefficient on the interaction term is positive and significant, implying that the impact of *IndepBoard* is lessened by the activities of HFT market makers. (For Zeros, for all intents and purposes, there is no discernable interaction effect.) In fact, generalizing across the four specifications,

the point estimate on the interaction term has roughly the same magnitude as the coefficient on *IndepBoard*, implying that the presence of HFT market makers effectively serves as a full substitute for board independence, at least with respect to liquidity. The bottom row of the table presents a formal test for the significance of the sum of the two coefficients. With the exception of the Amihud measure, the sum is not significant at conventional levels. By and large, therefore, in the presence of skilled market makers, there is little evidence that board independence has any impact on share liquidity. The results for the control variables in Table 4 are quite similar to those in Table 3, both in terms of signs and statistical significance.

[Table 4 here]

4.2.2 *CEO turnover*

We argued above that while managers concerned about private benefits may always have some incentive to hide information from the board, this effect can be exacerbated when managers are most concerned with their job security. For example, in periods that the board is engaged in negotiations for takeover and acquisitions, managers' jobs may be especially at risk, but these are not the only periods in which job risk is relevant. To capture such periods, we use an indicator variable, *CEO-TO*, to denote periods in which CEO turnover risk is especially high. The variable is set equal to 1 in any year that the CEO changes as a result of board action.

The results in Table 5 show that the interaction of *CEO-TO* with the proportion of independent directors is generally positive and is statistically significant for two liquidity measures, consistent with the hypothesis that tensions between the board and the CEO in these periods create information problems that offset the advantages otherwise provided by independent boards. The interaction terms are statistically significant for the Roll and the effective bid-ask spread. For both of these measures, the coefficient estimate on the interaction term has a higher magnitude than that on *IndepBoard*, indicating that the impediments to information sharing during CEO turnover periods actually reverse the direct beneficial effects of board independence. The bottom row of the table presents results for the sum of the coefficients on *IndepBoard* and *IndepBoard* $\times$ *CEO-TO*. The evidence here is

mixed, with a negative and significant sum for Amihud's measure, a positive and significant sum for the bid-ask spread, and insignificant results for the other two measures. On balance, across the four measure of liquidity, the point estimates provide strong evidence that an independent board enhances stock liquidity in normal times, but provide no evidence of such enhancement during periods of board-driven CEO turnover.

[Table 5 here]

4.2.3 *CEO bargaining power*

In the previous section, we established that market makers and other traders adjust their willingness to trade shares when they consider possibly tense negotiations between the board and CEO. CEO and board relations are also affected by the relative bargaining power between the two parties. CEO bargaining power may itself affect negotiations between these parties and whether CEOs choose to compete with the board to capture value or decide instead to cooperatively pursue value creation.

The CEO's bargaining power derives from his or her perceived ability compared to alternative managers that the firm might be able to hire. The CEO of a well-performing firm has greater bargaining power and is better positioned to negotiate with the board. In this regard, Hermalin and Weisbach (1998) believe that the intensity with which the board monitors the CEO decreases with its estimate of the CEO's ability. We predict that the manager's ability to control information and fend off intrusive monitoring may be increased substantially by high performance in previous periods, the effect of which we measure by Tobin's Q. Thus, we interact lagged Tobin's Q with IndepBoard to estimate the extent to which CEO bargaining power mitigates the impact of board independence on liquidity.

The results are presented in Table 6. The coefficient on IndepBoard remains negative and highly significant in every regression, and the magnitudes of the effects remain nontrivial. However, the interaction term has a positive effect on all four measures of illiquidity and is significant at the 5% level or better for three of them (Amihud illiquidity, Roll's spread, and the bid-ask spread). The consistently positive coefficients imply that the benefits of an independent board are reduced when dealing with a CEO who has high

bargaining power.¹³ This result is consistent with the notion that communication and cooperation between CEOs and boards is important. We conclude that the public interprets high CEO bargaining power as a signal of potential difficulties in negotiation with the board, which can result in information loss and reduced liquidity.

[Table 6 here]

4.2.4 *Investor Protection*

Considerable research on corporate governance suggests that investor protection rights at the country level can reinforce the effectiveness of internal governance at the firm level (e.g., Claessens et al., 2000; Lombardo and Pagano, 2006). Investor protection influences the relationships among managers, market makers, shareholders, creditors, and investors. In this regard, Chung et al. (2012) and Eleswarapu and Venkataraman (2006) find that firms operating in countries with strong investor protection tend to have better corporate governance structures, implying that the effectiveness and importance of a board in improving liquidity can depend on the legal and regulatory environment. Specifically, independent directors at the firm level might work in combination with investor protection at the country level. On the other hand, strong legal institutions may make independent boards less important to investors and market makers.

Following Defond and Hung (2004) and Leuz, Nanda and Wysocki (2003), we measure the strength of investor protection as the mean of three law enforcement variables identified in Leuz et al. (2003) and La Porta et al. (1999). The three variables are based on assessments from risk rating agencies (Business International Corp. or International Country Risk) that attempt to capture: (1) the efficiency and integrity of the country's judicial system; (2) the tradition of law and order in the country; and (3) the degree of government corruption, indicating whether high-level officials are likely to demand special payments and whether low level officials generally expect illegal payments (in the form of bribes). The index ranges from 0 to 10, with higher scores representing stronger law enforcement institutions. The data are taken from Leuz et al. (2003), Table 2 (Panel B, Legal Enforcement, p. 516).

¹³ Because Tobin's Q is not an indicator variable, the sum of the coefficients on *IndepBoard* and *IndepBoard* × *LagTQ* is not a meaningful measure of the full effect of board independence on liquidity.

Table 7 reports regressions that include controls for investor protection, IP. In general, firms that trade in countries with greater legal protection exhibit greater liquidity: the coefficients on IP are negative using three of the four illiquidity measures, and significant at the 1% level in two of the specifications. The interaction terms between IP and IndepBoard always have the opposite sign as the coefficient on IndepBoard, implying that the impact of independent directors is attenuated when controls for investor protection are included.

[Table 7 here]

Table 8 reports regressions estimates using the full set of interaction variables. These results are consistent with the one-at-a-time interaction regressions. Board independence continues to have a positive direct effect on liquidity, but the interaction terms all serve to mitigate that relation, as in the previous estimates. 15 of the 16 interaction coefficients are positive, and uniformly so for the statistically significant coefficients. Interestingly, there is no dominant interaction variable. Each retains high significance for at least some measures of liquidity.

[Table 8 here]

Table 8 allows us to measure the economic impact of the interaction terms. We measure the extent to which each of those terms reduces the direct effect of board independence on each liquidity measure. The baseline direct impact of board independence is measured as the impact on each liquidity measure resulting from an increase of one (sample) standard deviation in board independence. This is calculated as the product of the coefficient on board independence from Table 8 and its sample standard deviation from Table 1. Next we look at the impact of the interaction terms in mitigating the impact of board independence. The economic significance of each interaction term is measured as the percentage reduction in the baseline impact of board independence when the respective interaction variable is increased by one (sample) standard deviation. That reduction is measured at the sample-average value of board independence. Therefore, the mitigation effect of each interaction variable is the product of the coefficient on its interaction term times the sample standard deviation of the interaction variable times the average value of board independence. That product is then expressed as a percent of the baseline impact of board independence. The results of these calculations appear in Table 9.

Table 9 shows that the interaction terms generally offset a significant portion of the benefit derived from independent boards, with several entries in the neighborhood of 25%. Skilled HFT market makers and investor protection laws have the greatest economic significance. The combined economic significance of these interaction terms is consistently high, with column sums ranging from 48% to 59.8%, indicating that more than half of the baseline economic impact of board independence on liquidity can be reversed by these interaction terms.

[Table 9 here]

Finally, Table 10 presents results excluding Japanese firms from the sample. Table 1, Panel B, shows that observations from Japan dominate the sample. However, as Table 10 shows, Japan does not determine the results. Estimates excluding Japan from the sample are largely the same as those when it is included. Liquidity generally rises with board independence (direct effect), but the interaction terms all tend to reduce the impact of board independence, precisely as we found in Table 8, which included Japanese firms.¹⁴

[Table 10 here]

5. Conclusion

We investigate the link between board composition and liquidity of a firm's shares using a sample of East Asia companies. We hypothesize that greater board independence will, at least in most circumstances, increase firm transparency, and therefore induce market makers to offer greater liquidity in the firm's shares. In fact, we find strong evidence that greater board independence significantly increases liquidity.

However, the impact of board independence is affected by several other variables that influence the flow of information from managers to the board and the public. For example, some brokerages are better equipped to process information independently, and when these brokers make a market in shares, the importance of board independence on liquidity is substantially reduced. In some cases, when board-management relations are strained, for example when the board is engaged in considerations to replace management, the flow of

¹⁴ Results for Japan alone are not shown, but they too are consistent with the results in Table 8.

information from management to the board may be impeded. This effect clearly can be exacerbated when the board is more independent of management, and we find that in these periods, the beneficial impact of independence on liquidity is impaired. Similarly, when CEOs have greater bargaining power relative to the board, they may be less forthcoming with information, which again reduces the impact of independence on liquidity. Finally, when legal environments that provide greater investor protections generally tend to increase liquidity directly, they also appear to dampen the beneficial impact of board independence.

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Table1: Summary Statistics**Panel A: Full Sample**

The sample used for the baseline tests contains 2,407 firm-year observations. The sample used for some regressions may be smaller number due to data availability for a subset of variables. The sample covers 2003 to 2013, as these are the years for which Compustat data are available for East Asian Companies. Accounting and segment data are from Compustat. CEO turnover and the stock inventory of HFT market makers are taken from annual reports. The measurements of illiquidity are: Roll spread from Roll (1984); Zero returns from Lesmond, Odgen and Trzcinka (1999); Amihud illiquidity from Amihud (2002); and effective bid-ask spread from Stoll (1978). Amihud illiquidity (Amihud) is calculated as the monthly average of absolute daily return divided by the firm's daily dollar volume. Effective bid-ask spread (bid-ask) is the difference between bid and ask prices divided by their average. Zero returns (Zeros) is the proportion of days with zero reported returns. Roll spread (Roll) is the serial covariance of price changes. Independent board (IndepBoard) is the percentage of independent directors on the board. Trading volume (Volume) is the mean of daily dollar trading volume. Leverage (Leverage) is debt divided by book assets, firm size (FS) is the natural logarithm of the book value of assets, and return volatility (Volatility) is the standard deviation of daily closing quote-midpoint returns. Developed Market Dummy is a dummy variable that equals 1 if the companies trade in highly developed markets (Japan, Hong Kong and Singapore). HFT market makers (HFT-MM) is a dummy variable for companies for which HFT market makers (specifically, Barclays Capital and Goldman Sachs) make a market in their shares. CEO turnover (CEO-TO) is a dummy variable for firm-year observations in which the CEO is replaced by the board. Investor protection (IP) is the mean score of three law enforcement identified in Leuz et al. (2003); La Porta et al. (1999). Lag Tobin's Q (Lag TQ) is the one year lag value of Tobin's Q, estimated as the book value of assets minus the book value of equity plus the market value of equity, scaled by the book value of assets.

Variable	Mean	Median	Standard Deviation	Min	25th percentile	75th percentile	Max
Amihud illiquidity	0.09856	0.00255	0.31299	0.00004	0.00091	0.09118	9.96000
Roll spread	0.35129	0.01163	1.33583	0.00012	0.00078	0.07632	10.11448
Zero returns	0.28913	0.12260	0.32622	0.00056	0.07692	0.36015	1.00000
Effective bid-ask spread	0.02394	0.00992	0.03965	0.00032	0.00207	0.03362	0.71888
Independent Board	0.38994	0.36400	0.20650	0.03200	0.22200	0.54600	1.00000
HFT-Market Maker	0.06730	0.00000	0.25059	0.00000	0.00000	0.00000	1.00000
CEO Turnover dummy	0.10178	0.00000	0.30243	0.00000	0.00000	0.00000	1.00000
Investor Protection	8.13946	8.93000	1.60860	2.88000	7.37000	9.17000	9.17000
Lag Tobin's Q	1.02290	0.77163	1.08778	-0.26423	0.51156	1.17123	14.45430
Leverage	0.23454	0.21420	0.17237	0.00086	0.09221	0.34935	0.92094
Log Firm Size	16.03512	16.00556	1.81264	7.82109	15.03995	17.02130	21.66666
Volume (\$ thousands)	4673.737	10.92708	15171.75	0.00000	0.96667	1323.788	32388.40
Volatility	0.02460	0.02109	0.03008	0.00093	0.01576	0.02775	0.70710
Developed Market Dummy	0.76983	1.00000	0.42102	0.00000	1.00000	1.00000	1.00000

Panel B: Summary statistics by country

	China		Hong Kong		Indonesia		Japan		Malaysia		Singapore		South Korea		Taiwan		Thailand	
Variable	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Amihud	0.06210	0.69306	0.02722	0.02906	0.04462	0.10126	0.18595	0.29576	0.00303	0.01243	0.08257	0.27033	0.00141	0.01771	3.96887	0.00000	0.03004	0.04981
Roll	0.04802	0.13186	0.02626	0.11301	0.09347	0.48565	0.12452	0.31867	0.02958	0.09390	0.03555	0.06481	0.10208	0.12848	0.12343	0.33936	0.06940	0.30906
Zeros	0.18988	0.21459	0.53804	0.30029	0.54379	0.28385	0.31381	0.37517	0.40277	0.34197	0.35025	0.28387	0.13677	0.20102	0.11398	0.02980	0.05088	0.05847
Bid-ask	0.01264	0.02731	0.05191	0.04280	0.05954	0.03376	0.0258 7	0.01625	0.00943	0.03123	0.06009	0.07476	0.00289	0.00365	0.00269	0.00198	0.36162	0.32690
IndepBoard	39.55000	8.43652	35.57683	11.43842	48.83521	15.89351	25.69050	16.50665	50.49349	14.9164	59.4953	18.2087	56.1543	12.6969	29.3807	13.1797	48.3140	15.59358
HFT-MM	0.10377	0.30568	0.23170	0.42451	0.08450	0.28012	0.07100	0.25695	0.02439	0.15488	0.00000	0.00000	0.1035	0.30230	0.05729	0.23300	0.00000	0.00000
CEO-TO	0.10377	0.30568	0.12195	0.32924	0.05633	0.23221	0.12900	0.33536	0.10569	0.30869	0.06267	0.24300	0.08783	0.28353	0.05208	0.22277	0.12500	0.33333
Lag TQ	1.20391	0.14123	0.97871	0.96817	2.43382	2.94107	0.79978	0.72258	1.29001	1.48005	1.06927	0.78112	0.96656	1.05297	1.17666	0.97220	1.38414	1.48846
Leverage	0.24146	0.17363	0.18887	0.15696	0.15654	0.16228	0.23645	0.17891	0.22476	0.15226	0.21868	0.14749	0.28770	0.18366	0.21139	0.16411	0.26554	0.16724
Log FS	18.37355	2.02906	19.72428	1.98740	24.56930	1.33957	20.72639	1.61887	16.64680	1.81270	15.4532	2.01016	23.1508	1.60495	18.9089	1.54462	19.4799	1.48188
Volume	1824.42	3208.66	3438.72	1854.32	2890.45	1780.71	2487.13	1893.46	6824.32	2267.91	2826.29	3146.04	5894.53	1752.01	9281.54	1111.85	2924.77	2455.81
Volatility	0.02855	0.03271	0.03351	0.02502	0.04081	0.05035	0.02091	0.03103	0.01399	0.01987	0.03029	0.02989	0.02742	0.02975	0.02237	0.00601	0.02671	0.02537
N (firm-years)	212		82		71		1,000		118		367		296		192		64	

Table 2: Correlation matrix for baseline specification variables.

Definitions of variables are in Table 1. Pearson correlations are reported below the main diagonal and Spearman correlations are reported above the diagonal.

	Amihud	Roll	Zeros	Bid-Ask	IndepBoard	HFT-MM	CEO-TO	Lag TQ	IP	Leverage	Log FS	Volume	Volatility
Amihud		0.2807***	0.1927***	0.5344***	-0.2242***	-0.0035	0.0749**	-0.0183	0.6120***	-0.0181	0.1110***	-0.8745***	0.0900**
Roll	0.0380		0.4223***	0.7000***	0.0448	-0.0955***	0.0385	-0.0787**	0.1130***	-0.0074	-0.2584***	-0.1634***	0.5628***
Zeros	-0.0125	0.0072		0.4482***	-0.2234***	-0.1674***	-0.0072	-0.0404	-0.3347***	0.1078***	-0.3311***	0.3447***	0.2382***
Bid-Ask	0.1095***	0.3817***	0.2564***		0.0388	-0.1409***	0.0663**	-0.0879***	0.2163***	0.0319	-0.1995***	-0.4426***	0.4561***
IndepBoard	-0.1266***	-0.0685***	-0.0114	-0.0098		0.0070	-0.0230	0.0355	-0.4517***	-0.1380***	-0.1803***	0.2827***	0.1384***
HFT-MM	-0.0077	0.0186	0.0091	-0.0920***	0.0247		-0.0070	-0.0470	0.0914***	-0.0504	0.1833***	-0.0294	-0.1288***
CEO-TO	0.0213	0.0027	0.0012	0.0225	-0.0195	0.0193		-0.0241	0.0710**	0.0181	0.0420	-0.0906***	0.0067
Lag TQ	0.0433**	-0.0261	0.0480**	-0.0319	0.1004***	0.1096***	-0.0496**		-0.1592***	0.0510	-0.2481***	0.0134	-0.0224
IP	0.1716***	0.1881***	0.0568***	0.0931***	-0.3231***	-0.0146	0.0412**	-0.1881***		0.0163	-0.0083	-0.7002***	-0.1816***
Leverage	-0.0551***	-0.0384	-0.0137	-0.0242	-0.0844***	-0.0617***	0.0247	-0.1007***	-0.0268		0.0219	0.0863***	0.0571*
Log FS	-0.0115	-0.1704***	-0.0846***	-0.2319***	-0.1309***	0.1884***	0.0725***	-0.1992***	-0.3895***	0.1049***		-0.1615***	-0.0271
Volume	-0.1325***	-0.0033	-0.1413***	-0.1880***	0.0204	0.0014	-0.0141	-0.0082	0.0240	-0.0211	-0.0521**		0.0894***
Volatility	0.0890***	0.2227***	-0.0163	0.4743***	0.0630***	-0.0198	0.0113	0.0977***	-0.0805***	-0.0082	-0.1001***	-0.256	

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively

Table 3: Liquidity and board independence

This table presents the results for the relation between board independence and liquidity. The descriptions for all variables are given in Table1. This table presents the estimation results of the dynamic GMM model. Time (year) dummies are included in the regressions. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	0.21773*** (3.74)	0.80068*** (14.87)	0.59100*** (11.13)	0.15510*** (8.04)
IndepBoard	-0.00260** (-2.41)	-0.02086*** (-3.31)	-0.00123*** (-3.33)	-0.00020*** (-3.78)
Leverage	0.04233 (0.24)	-0.02512 (-0.03)	0.05817 (0.79)	-0.06193*** (-3.91)
Log FS	-0.01053 (-0.69)	-0.18266** (-2.53)	-0.02590*** (-3.41)	-0.00663*** (-4.55)
Volume	-0.00210*** (-3.37)	0.01093 (1.25)	0.00012 (0.56)	-0.00034*** (-5.80)
Volatility	0.95569 (1.50)	-3.11766 (-0.69)	0.58685* (1.73)	1.68248*** (12.35)
Market Dummy	0.00964 (0.17)	-0.42207 (-1.61)	-0.05832** (-2.03)	-0.00543 (-0.88)
AR (1) test (p-value)	(0.010)	(0.001)	(0.000)	(0.036)
AR (2) test (p-value)	(0.980)	(0.891)	(0.465)	(0.755)
Hansen-J test (p-value)	(0.297)	(0.141)	(0.454)	(0.516)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 4: Interaction effects with HFT market makers

This table presents results for the relation between board independence and liquidity including interaction terms for when HFT market makers make markets in the stocks. The descriptions for all of the variables are given in Table1. This table presents the estimation results of the dynamic GMM model. Time (year) dummies are included in the regressions. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	0.02031 (0.45)	0.73828*** (13.65)	0.59598*** (12.97)	0.15550*** (8.41)
IndepBoard	-0.00447*** (-3.98)	-0.02669*** (-4.74)	-0.00146*** (-3.74)	-0.00017** (-2.49)
IndepBoard × HFT-MM	0.00963*** (5.31)	0.02388*** (4.15)	-0.00001 (-0.00)	0.00024** (2.58)
HFT-MM	-0.44764*** (-5.07)	-0.32657 (-1.23)	0.11056 (1.16)	0.00087 (0.14)
Leverage	-0.18961 (-1.41)	-0.09060 (-0.12)	-0.03638 (-0.71)	-0.06344*** (-4.18)
Log FS	-0.00159 (-0.11)	-0.26448*** (-3.58)	-0.03484*** (-4.58)	-0.00645*** (-4.63)
Volume	-0.00358*** (-4.77)	0.01555* (1.72)	-0.00019 (-0.88)	-0.00034*** (-5.98)
Volatility	1.96218*** (3.46)	-1.68735 (-0.39)	-0.39661** (-2.00)	1.66625*** (13.63)
Market Dummy	0.05321 (0.99)	-0.56613* (-1.92)	-0.08651*** (-3.26)	-0.00443 (-0.75)
AR (1) test (p-value)	(0.037)	(0.001)	(0.000)	(0.039)
AR (2) test (p-value)	(0.834)	(0.842)	(0.255)	(0.743)
Hansen-J test (p-value)	(0.892)	(0.510)	(0.585)	(0.985)
Sum of coefficients on IndepBoard and IndepBoard × HFT-MM	.00516*** (3.45)	-.00281* (1.84)	-.00147 (1.09)	.00007 (1.73)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 5: Interaction effects with board-instigated CEO turnover

This table presents results for the relation between board independence and liquidity including interaction terms for periods of board-instigated CEO turnover. The descriptions of all of the variables are given in Table1. This table presents the estimation results of the dynamic GMM model. Time (year) dummies are included in the regressions. *t*-statistics are reported in parentheses. R1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	0.21519*** (3.97)	0.87919*** (20.71)	0.61593*** (8.47)	0.24219*** (10.36)
IndepBoard	-0.00322*** (-2.98)	-0.00934** (-2.12)	-0.00098** (-2.34)	-0.00002 (-0.45)
IndepBoard × CEO-TO	-0.00158 (-1.41)	0.01967** (2.36)	0.00067 (0.79)	0.00040** (2.51)
CEO-TO	0.09068 (1.38)	-0.76969** (-2.18)	-0.02295 (-0.59)	-0.02165** (-2.53)
Leverage	0.06160 (0.42)	1.11951 (1.27)	0.05199 (0.59)	-0.04790*** (-3.74)
Log FS	-0.00893 (-0.55)	-0.07639 (-1.28)	-0.02597*** (-3.12)	-0.00288** (-2.50)
Volume	-0.00201** (-2.57)	0.00496 (0.55)	0.00012 (0.56)	-0.00029*** (-6.10)
Volatility	0.85985* (1.71)	3.12512 (0.81)	-0.18205 (-0.55)	1.51530*** (13.51)
Market Dummy	-0.00159 (-0.03)	-0.11701 (-0.57)	-0.05095* (-1.69)	0.00854* (1.74)
AR (1) test (p-value)	(0.009)	(0.002)	(0.000)	(0.044)
AR (2) test (p-value)	(0.924)	(0.649)	(0.317)	(0.660)
Hansen-J test (p-value)	(0.913)	(0.162)	(0.824)	(0.378)
Sum of coefficients on IndepBoard and IndepBoard × CEO-TO	-.00480*** (-2.65)	.01033 (0.93)	-.00031 (-0.38)	.00038*** (2.40)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 6: Interaction effects with Tobin's Q

This table presents results for the relation between board independence and liquidity including interaction terms with Tobin's Q (TQ). The descriptions for all of the variables listed following are given in Table1. This table presents the estimation results for the dynamic GMM model. Time (year) dummies are included in the regressions. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	-0.12394*** (-5.14)	0.62536*** (7.62)	0.60768*** (14.19)	0.16248*** (9.57)
IndepBoard	-0.00504*** (-7.18)	-0.02564*** (-2.73)	-0.00203*** (-5.41)	-0.00022** (-2.56)
IndepBoard × LagTQ	0.00055*** (2.75)	0.01086** (2.16)	0.00018* (1.75)	0.00009** (2.35)
LagTQ	-0.01953 (-1.58)	-1.28685*** (-3.26)	-0.01098 (-1.25)	-0.01151*** (-3.91)
Leverage	-0.25351*** (-2.79)	-0.11856 (-0.09)	-0.03685 (-0.52)	-0.11617*** (-6.31)
Log FS	-0.00568 (-0.49)	-0.43303*** (-3.27)	-0.02829*** (-3.96)	-0.00885*** (-5.84)
Volume	-0.00373*** (-6.45)	0.00933 (0.91)	0.00002 (0.09)	-0.00029*** (-5.59)
Volatility	1.22344** (2.38)	23.72447*** (6.83)	0.77888** (2.45)	1.49441*** (11.87)
Market Dummy	0.03494 (0.92)	-0.99968* (-1.73)	-0.07991*** (-2.91)	-0.01584** (-2.25)
AR (1) test (p-value)	(0.035)	(0.003)	(0.000)	(0.034)
AR (2) test (p-value)	(0.582)	(0.700)	(0.416)	(0.781)
Hansen-J test (p-value)	(0.163)	(0.116)	(0.736)	(0.791)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 7: Interaction effects with degree of investor protection

This table presents results for the relation between board independence and liquidity including interaction terms for degree of investor protection (IP) provided by legal system in the firm's home country. The descriptions for all of the variables listed following are given in Table1. This table presents the estimation results of the dynamic GMM model. Time (year) dummies are included in the regressions. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	0.18902*** (3.70)	0.65817*** (12.62)	0.54352*** (10.49)	0.18178*** (8.68)
IndepBoard	-0.02024*** (-3.02)	-0.08008** (-1.89)	-0.00933*** (-3.99)	0.00291 (1.63)
IndepBoard × IP	0.00206*** (2.63)	0.00655 (1.30)	0.00094*** (3.29)	-0.00036** (-1.84)
IP	-0.05905* (-1.72)	-1.24445*** (-2.84)	-0.04895*** (-3.22)	0.01125 (1.32)
Leverage	0.00710 (0.05)	-0.64501 (-1.00)	-0.08174 (-1.37)	-0.12452*** (-4.71)
Log FS	0.00519 (0.44)	-0.12799* (-1.67)	-0.02206** (-2.23)	-0.00473** (-2.32)
Volume	-0.00203*** (-3.38)	0.00301 (0.35)	-0.00035 (-1.43)	-0.00027*** (-4.83)
Volatility	1.48601*** (2.79)	-0.49994 (-0.14)	-0.15238*** (-2.79)	1.46888*** (12.95)
Market Dummy	-0.10160** (-1.98)	3.64897** (2.02)	-0.04974* (-1.66)	0.01896 (1.60)
AR (1) test (p-value)	(0.014)	(0.001)	(0.003)	(0.040)
AR (2) test (p-value)	(0.916)	(0.801)	(0.308)	(0.745)
Hansen-J test (p-value)	(0.154)	(0.116)	(0.117)	(0.295)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 8: Interaction effects with full set of mediating variables

This table presents dynamic GMM estimates for the relation between board independence and liquidity including interaction terms for the full set of mediating variables. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	-0.06552*** (-3.48)	0.42560*** (8.42)	0.51477*** (20.74)	0.25394*** (13.44)
IndepBoard	-0.01266*** (-2.72)	-0.05325 (-0.67)	-0.00929*** (-3.04)	-0.00316** (-2.57)
IndepBoard × HFT-MM	0.00617*** (8.59)	0.03392*** (4.23)	0.00206*** (4.94)	0.00123*** (4.71)
IndepBoard × CEO-TO	-0.00006 (-0.33)	0.00619*** (2.69)	0.00084** (2.00)	-0.00006 (-0.89)
Indepboard × LagTQ	0.00083*** (7.47)	0.00231 (1.04)	0.00019** (2.37)	0.00005 (0.71)
IndepBoard × IP	0.00098** (1.93)	0.00170 (0.19)	0.00086** (2.55)	0.00029** (2.14)
HFT-MM	-0.28384*** (-8.13)	-1.06793* (-1.95)	-0.12559 (-1.39)	-0.03079** (-2.09)
CEO-TO	-0.00054 (-0.04)	-0.60565*** (-4.60)	-0.03448* (-1.77)	0.00844** (2.30)
LagTQ	-0.03880*** (-7.53)	-0.11038 (-0.67)	-0.02589*** (-7.59)	-0.02184*** (-3.45)
IP	0.06861** (1.83)	-1.27611 (-1.40)	-0.09637*** (-3.35)	-0.02666* (-1.91)
Leverage	-0.47289*** (-9.12)	0.16150 (0.27)	-0.19861*** (-3.78)	-0.10200*** (-6.77)
Log FS	0.00590 (0.70)	-0.19577*** (-3.83)	-0.02313*** (-2.77)	-0.01181*** (-4.10)
Volume	-0.00262*** (-4.48)	0.00456 (0.49)	-0.00060*** (-2.71)	-0.00026*** (-3.73)
Volatility	1.36684*** (5.45)	24.50765*** (8.73)	-0.18595** (-2.04)	0.59789*** (5.05)
Market Dummy	-0.31992*** (-3.19)	4.92878 (1.41)	0.04977 (0.53)	-0.00057 (-0.01)
AR (1) test (p-value)	(0.044)	(0.003)	(0.000)	(0.011)
AR (2) test (p-value)	(0.585)	(0.297)	(0.403)	(0.602)
Hansen-J test (p-value)	(0.350)	(0.887)	(0.795)	(0.894)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).

Table 9: Economic significance of interaction terms

The baseline economic significance of board independence is the impact on each liquidity measure resulting from an increase of one standard deviation in board independence. This is calculated as the product of the coefficient on board independence from Table 8 and its sample standard deviation from Table 1. The economic significance of each interaction term with board independence is the percentage reduction in the baseline impact of board independence when the respective interaction variable is increased by one (sample) standard deviation. The reduction is measured at the sample-average value of board independence. Therefore, the mitigation effect of each interaction variable is the product of the coefficient on its interaction term times the sample standard deviation of the variable times the average value of board independence. That product is expressed as a percent of the baseline impact of board independence.

Liquidity measure:	Amihud	Roll	Zeros	Bid-Ask spread
<u>Interaction term</u>				
HFT market makers	-23.1%	-30.1%	-10.5%	-18.4%
CEO turnover	0.3%	-6.6%	-5.2%	1.1%
Tobin's Q (lagged)	-13.5%	-8.9%	-4.2%	-3.2%
Investor protection laws	-23.5%	-9.7%	-28.1%	-27.9%

0.627257

HFT Amih $0.00617 * 0.38994 * 0.25059 = 0.00060290$

Table 10: Interaction effects with full set of mediating variables for non-Japanese firms.

This table presents dynamic GMM estimates for the relation between board independence and liquidity including interaction terms for the full set of mediating variables for Hong Kong, Singaporean, Indonesian, Malaysian, Taiwanese, Chinese, Thailand and South Korean companies. *t*-statistics are reported in parentheses. AR1 and AR2 are tests for first-order and second-order serial correlation in the first-differenced residuals. The Hansen test of over-identification is under the null that all instruments are valid.

	Amihud	Roll	Zeros	Bid-Ask
Dependent variable (<i>t</i> -1)	0.04647*** (24.61)	0.38387*** (13.17)	0.60305*** (9.74)	0.60071*** (4.62)
IndepBoard	-0.00464*** (-3.24)	-0.02596*** (-4.63)	-0.01463*** (-4.94)	-0.00325 (-1.34)
IndepBoard × HFT-MM	0.00123** (1.93)	0.01925*** (6.24)	0.00714*** (6.88)	0.00021 (1.01)
IndepBoard × CEO-TO	0.00030 (0.35)	0.00683* (1.92)	0.00359* (1.66)	-0.00004 (-0.04)
Indepboard × LagTQ	0.00017*** (2.85)	-0.00043 (-0.84)	-0.00004 (-0.29)	0.00050* (1.96)
IndepBoard × IP	0.00057*** (3.45)	0.00303*** (4.37)	0.00142*** (4.71)	0.00029 (1.23)
HFT-MM	-0.03878 (-1.61)	-0.71320*** (-5.71)	-0.07925 (-1.46)	-0.011915 (-0.74)
CEO-TO	-0.02262 (-0.67)	-0.27899* (-1.93)	-0.12183 (-1.48)	-0.00710 (-0.19)
LagTQ	-0.01050** (-2.37)	0.02129 (0.70)	-0.00741 (-0.76)	-0.01577 (-1.35)
IP	-0.02353*** (-3.19)	-0.11073*** (-3.87)	-0.04562** (-2.39)	-0.00593 (-0.61)
Leverage	-0.01646 (-0.44)	0.14562** (2.45)	0.15156* (1.77)	-0.02256 (-0.84)
Log FS	0.00093 (0.24)	0.03232*** (3.67)	0.02326*** (3.20)	0.00340* (1.79)
Volume	-0.00641** (-2.27)	0.00008 (1.01)	-0.02056*** (-5.78)	-0.00449*** (-2.79)
Volatility	0.51749* (1.98)	0.37507 (0.35)	-0.00203 (0.00)	0.23599 (0.55)
AR (1) test (<i>p</i> -value)	(0.041)	(0.045)	(0.033)	(0.007)
AR (2) test (<i>p</i> -value)	(0.253)	(0.524)	(0.881)	(0.257)
Hansen-J test (<i>p</i> -value)	(0.938)	(0.885)	(0.589)	(0.872)

***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively (two-tailed tests).